AI-Assisted Kidney Stone Detection & Classification Report

Research Examination ID: 511A05CB | Date: 2026-10-28 11:01:50

1. Image & Study Metadata

Parameter	Value
Examination ID	511A05CB
Modality	CT Scan
Image Format	JPEG
DICOM Calibrated	No (Standard 2D Image)

2. Individual Deep Learning Classification Models

Model Architecture	Status	Predicted Class	Confidence	Stone Prob.	Non-Stone Prob.	Latency
MOBILENET V2	Trained & Loaded	**Stone**	85.7%	85.7%	14.3%	866.41 ms
RESNET50	Trained & Loaded	**Stone**	99.7%	99.7%	0.3%	297.04 ms
EFFICIENTNET B0	Trained & Loaded	**Stone**	99.1%	99.1%	0.9%	402.52 ms

3. Multi-Model Ensemble Analysis

Ensemble Decision	**Stone** (94.8% confidence)
Stone Probability	94.8%
Non-Stone Probability	5.1%
Contributing Classifiers	mobilenet_v2, resnet50, efficientnet_b0
Normalized Model Weights	{'mobilenet_v2': 0.3333, 'resnet50': 0.3333, 'efficientnet_b0': 0.3333}
Model Disagreement	Full Consensus

4. YOLOv8 Object Detection (Localization)

Detected 5 region(s) using YOLOv8.

5. Physical Measurements & Calibrations

Status: Uncalibrated 2D ROI: 30 x 25 pixels

• Physical millimeter scale unavailable: Standard PNG/JPEG image does not retain DICOM PixelSpacing metadata.

• Reporting pixel dimensions only. Do NOT make clinical decisions based on uncalibrated pixel sizes.

MANDATORY MEDICAL & RESEARCH DISCLAIMER:
This report was generated by an academic research artificial intelligence system. The classification outputs, probability scores, and ensemble results are algorithmic estimates and do NOT constitute a medical diagnosis, clinical radiologist verification, or treatment recommendation. Model confidence

must not be interpreted as an absolute probability of clinical disease. Clinical diagnosis requires thorough review by a qualified radiologist or physician.